

01_standardization_params

```
source("R/00_setup.R")
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(haven)

sub      <- readRDS("WVS_subset_40countries_W4toW7.rds")
coords  <- readRDS("country_wave_coords.rds")

#' Column names in the WVS trend file vary in case across releases, so resolve
#' the locked instrument against whatever is actually present and fail loudly if
#' any component is missing.

resolve <- function(want, have) unname(setNames(have, toupper(have))[toupper(want)])
nms     <- names(sub)
item_cols <- resolve(ITEMS, nms)
stopifnot(!any(is.na(item_cols)))

#' Respondent-level matrix of the thirteen components, WVS negatives → NA.

X <- vapply(item_cols,
```

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        function(cc) { v <- as.numeric(zap_labels(sub[[cc]])); v[v < 0] <- NA; v },
        numeric(nrow(sub)))
colnames(X) <- ITEMS

#' The parameters themselves: pooled over the entire respondent sample, computed
#' once, and unweighted. Weighting enters later, at aggregation to national
#' means, not here.

mu <- apply(X, 2, mean, na.rm = TRUE)
sg <- apply(X, 2, sd, na.rm = TRUE)

params <- tibble(item = ITEMS, dim = DIM_OF[ITEMS], sign = SIGNS[ITEMS],
                 mu = mu[ITEMS], sd = sg[ITEMS])
write_csv(params, "standardization_params.csv")
print(params)

```

```

# A tibble: 13 x 5
  item dim  sign  mu  sd
  <chr> <chr> <dbl> <dbl> <dbl>
1 F063 Dim1  -1 7.67 3.06
2 A029 Dim1   1 0.497 0.500
3 A039 Dim1   1 0.365 0.482
4 A040 Dim1  -1 0.410 0.492
5 A042 Dim1  -1 0.379 0.485
6 F120 Dim1   1 3.23 2.81
7 G006 Dim1   1 1.55 0.735
8 E018 Dim1   1 1.55 0.731
9 Y002 Dim2   1 1.78 0.623
10 A008 Dim2  -1 1.89 0.727
11 F118 Dim2   1 3.58 3.21
12 E025 Dim2  -1 2.20 0.814
13 A165 Dim2  -1 1.72 0.447

```

```

#' ## Verification
#'
#' Rebuild the saved coordinates using only the exported parameters. This
#' repeats the full ground-truth aggregation: z-score, apply the orientation
#' sign, restrict to the constant within-country indicator base, take
#' survey-weighted national means per item, and average items within each
#' dimension.

```

```

Z <- sweep(sweep(X, 2, params$mu, `~`), 2, params$sd, `/~`)
Z <- sweep(Z, 2, params$sign, `*`)

wave <- as.numeric(sub[[resolve("S002VS", nms)]]])
w <- as.numeric(sub[[resolve("S017", nms)]]]); w[is.na(w) | w <= 0] <- 1
df <- data.frame(iso = sub$iso3, wave, w, Z, check.names = FALSE)

#' The constant within-country indicator base: keep an item for a country only if
#' it is more than half fielded in *all* of that country's waves. Applying the
#' same base to every wave is what keeps a country's coordinate comparable across
#' its own trajectory.

in_base <- df %>%
  group_by(iso, wave) %>%
  summarise(across(all_of(ITEMS), ~ mean(!is.na(.x)) > 0.5), .groups = "drop") %>%
  group_by(iso) %>%
  summarise(across(all_of(ITEMS), ~ all(.x)), .groups = "drop") %>%
  pivot_longer(all_of(ITEMS), names_to = "item", values_to = "keep")

wmean <- function(x, w) { ok <- !is.na(x); if (!any(ok)) NA_real_ else sum(x[ok] * w[ok]) / sum(w[ok]) }

repro <- df %>%
  pivot_longer(all_of(ITEMS), names_to = "item", values_to = "z") %>%
  left_join(in_base, by = c("iso", "item")) %>% filter(keep) %>%
  group_by(iso, wave, item) %>% summarise(m = wmean(z, w), .groups = "drop") %>%
  mutate(dim = DIM_OF[item]) %>%
  group_by(iso, wave, dim) %>% summarise(score = mean(m, na.rm = TRUE), .groups = "drop") %>%
  pivot_wider(names_from = dim, values_from = score) %>%
  rename(Dim1r = Dim1, Dim2r = Dim2)

chk <- coords %>% select(iso, wave, Dim1, Dim2) %>% inner_join(repro, by = c("iso", "wave"))
cat("max coord diff vs saved:",
    format(max(abs(chk$Dim1 - chk$Dim1r), abs(chk$Dim2 - chk$Dim2r)), scientific = TRUE), "\n")

```

max coord diff vs saved: 0e+00

```

#' Note for the methods: this operationalizes the Inglehart-Welzel dimensions as
#' an equal-weight average of z-scored component items, not as the published
#' `tradrat5` / `survsself` factor scores. The composite is a deliberate proxy,
#' chosen because the identical rule can be applied to model outputs.

```